



200 North Lake Street • Cadillac, Michigan 49601
231.775.0181 • fax 231.775.8755
www.cadillac-mi.net

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JUL 07 2014

7/2/2014

Mr. Dave Schipper
Department of Environmental Quality
350 Ottawa N.W.
Grand Rapids, MI 49503

DEQ
WATER RESOURCES DIVISION
GRAND RAPIDS

Subject: Amendment to Existing RMP

Dear Mr. Schipper:

The City of Cadillac would like to add a method to Section III (Land Application Plan) Part B-4 of our RMP whereas we can surface apply biosolids. We would like to have this option, if the sub-surface injection cannot be carried out. We meet the criteria for "Exceptional Quality" Biosolid as listed in Part 24;

- (i) Pollutant ceiling concentrations in R 323.2409(5)(a)
- (ii) Pollutant concentrations in R 323.2409(5)(c)
- (iii) Vector attraction reduction options in R 323.2415(4) (a); *we had 54.11% volatile reduction in 2013 with primary digester heated to average of 96.20 °F using external heat exchanger and retention time of 22.2 days.* And class A pathogen reduction alternatives in R 323.2414(2)(a); *the density of salmonella sp. bacteria in the biosolids was <3 MPN per 4 grams of total solids (dry weight basis) for 2013.*

The following analytical data represents the average contents of biosolids in our most recent analysis. Please find the attached documents for the complete list of the variables analyzed.

Recent biosolids analysis in mg/kg (dry weight basis)

Name	Concentration (mg/kg) Dry Weight	Pollutant Concentrations (Table 3, 40 CFR 503.13) (monthly average)	Ceiling Concentrations* (Table 1, 40 CFR 503.13) (daily maximum)
Arsenic	15.3	41 mg/kg	75 mg/kg
Cadmium	2.7	39 mg/kg	85 mg/kg
Copper	706	1500 mg/kg	4300 mg/kg
Lead	<50	300 mg/kg	840 mg/kg
Mercury	0.72	17 mg/kg	57 mg/kg
Molybdenum	23	N/A**	75 mg/kg
Nickel	18.6	420 mg/kg	420 mg/kg
Selenium	4.2	100 mg/kg	100 mg/kg
Zinc	1060	2800 mg/kg	7500 mg/kg
Nitrogen Concentration		N/A	N/A

* Biosolids may not be land applied if any pollutant exceeds these values.

** EPA has temporarily removed molybdenum limits from Table 2, Table 3 and Table 4

If you have any questions, please do not hesitate to contact me.

Sincerely,



Hamid Rahbarnoohi, Water Resources Supervisor

City of Cadillac

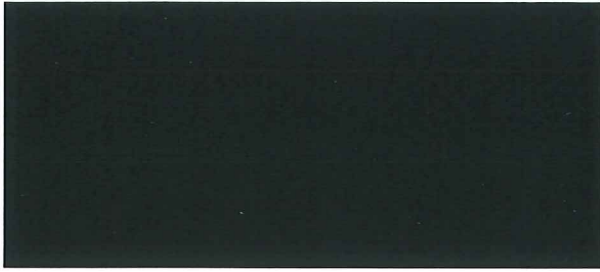
200 N. Lake Street

Email: hrahbarnoohi@cadillac-mi.net; Phone: 231-775-2841; Fax: 231-775-2876

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**DEQ
WATER RESOURCES DIVISION
GRAND RAPIDS**



Testing • Research • Consulting

Multiple Analysis Report

Sample: 268372

July 24, 2013

Ms. Amy Vail
City of Cadillac
200 N. Lake St.
Cadillac, MI 49601

RE: PWS ID#: Unavailable
City of Cadillac
Digester #4
1121 Plett Rd.
Cadillac, MI 49601

Dear Ms. Vail:

The following are the result(s) of the test(s) performed on the sample(s) received at HML, Inc. at 10:26 AM, 04/16/2013, and collected at 2:00 PM, 04/15/2013:

<u>TEST - METHOD</u>	<u>RESULT</u>	<u>MDL*</u>	<u>Date Complete</u>
Fecal Coliform-SM9221E (MPN)	1,160 mpn/g dry	1 MPN/g dry	04/19/2013
Enteric Virus-EPA 625/R-92/013	<1 pfu/4g dry	1 pfu/4g dry	07/23/2013
Solids, total-SM 2540G	2.483%	0.005%	05/13/2013
Salmonella-EPA 1682	<3 MPN/4 g dry	3 MPN/4 g dry	04/18/2013
Helminth Ova-EPA/625/R-92/013	<1 ovum/4g dry	1 ovum/4 g dry	07/19/2013

*Minimum Detection Level

This testing was completed by J.B. and N.R. Please feel free to contact us if we can be of further service to you.

Sincerely,

Donald A. Hendrickson, Ph.D.
President - Microbiologist
Chemistry Lab #C-18-01
Microbiological Lab #M-18-03

DAH/skp

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DEQ
WATER RESOURCES DIVISION
GRAND RAPIDS

LABORATORY DETAIL REPORT

Client: **City of Cadillac, Water Resources**

KAR Project No. : 402028

Date Reported : 04/15/14

Attest: 
David R. Alkema, Lab Manager

JUL 07 2014

Project

Description: Analysis of one Spring Biosolids sample.

DEQ
WATER RESOURCES DIVISION
GRAND RAPIDS

Sample ID : **"Digester #4"**

Sampled By : **DC of City of Cadillac, Water Resources**

Date Received : **04/01/14**

Sample Date : **03/31/14**

Sample Type : **sludge**

Sample Time : **1300**

KAR Sample No. : **402028-01**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, Hg	Completed		EPA 7471A	04/14/14	BPK	
Prep, metals	Completed		EPA 200.7/8	04/03/14	ZZW	
Aluminum, total	2840	mg/kg total solids	EPA 6010B	04/11/14	JHB	Detection limit = 46
Arsenic, total	15.3	mg/kg total solids	EPA 200.8	04/08/14	ZZW	Detection limit = 1
Cadmium, total	2.7	mg/kg total solids	EPA 200.8	04/08/14	ZZW	Detection limit = 0.2
Calcium, total	18,900	mg/kg total solids	EPA 6010B	04/11/14	JHB	Detection limit = 46
Chromium, total	81.8	mg/kg total solids	EPA 6010B	04/11/14	JHB	Detection limit = 4.6
Copper, total	706	mg/kg total solids	EPA 6010B	04/11/14	JHB	Detection limit = 9.1
Lead, total, by ICP	<50	mg/kg total solids	EPA 200.8	04/15/14	ZZW	
Magnesium, total	3040	mg/kg total solids	EPA 6010B	04/11/14	JHB	Detection limit = 46
Mercury, total	0.72	mg/kg total solids	EPA 7471A	04/14/14	BPK	Detection limit = 0.05
Molybdenum, total	23.0	mg/kg total solids	EPA 6010B	04/11/14	JHB	Detection limit = 9.1
Nickel, total	18.6	mg/kg total solids	EPA 6010B	04/11/14	JHB	Detection limit = 9.1
Potassium, total	90.8	mg/kg total solids	EPA 6010B	04/11/14	JHB	Detection limit = 46
Selenium, total	4.2	mg/kg total solids	EPA 200.8	04/08/14	ZZW	Detection limit = 1
Silver, total	<4.6	mg/kg total solids	EPA 6010B	04/14/14	JHB	
Sodium, total	4630	mg/kg total solids	EPA 6010B	04/11/14	JHB	Detection limit = 46
Zinc, total	1060	mg/kg total solids	EPA 6010B	04/11/14	JHB	Detection limit = 9.1
TCLP extraction	Completed		EPA 1311	04/02/14	JWW	
TCLP report	See comment			04/15/14	SRM	This material does not exhibit the Toxicity Characteristic with respect to the requested TC parameters.
ZHE extraction	Completed		EPA 1311	04/02/14	JWW	
Chloride	6780	mg/kg total solids	EPA 300.0A	04/04/14	ALK	Detection limit = 190
Nitrogen, ammonia	36,100	mg/kg total solids	EPA 350.1	04/02/14	JWW	Detection limit = 1500
Nitrogen, nitrate	<37	mg/kg total solids	EPA 353.2	04/03/14	ALK	
Nitrogen, total kjeldahl	81,100	mg/kg total solids	EPA 351.2	04/07/14	ALK	Detection limit = 2600
Phosphorus, total (as P)	41,800	mg/kg total solids	SM 4500-P B5,E	04/08/14	JHB	Detection limit = 3000
Sulfate	2110	mg/kg total solids	EPA 300.0A	04/04/14	ALK	Detection limit = 370
Waste pH	7.7	S.U.	EPA 9045C	04/09/14	AMG	
Solids, total	2.74	% by weight	EPA 160.3	04/03/14	AMG	Detection limit = 0.01
Solids, volatile	57.70	% of total solids	EPA 160.4	04/03/14	AMG	Detection limit = 0.01
Prior. Poll. volatiles	See below		EPA 8260	04/02/14	JAR	
Prep, VOA	Completed		EPA 5030	04/02/14	JAR	
1,1,1-Trichloroethane	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
1,1,2,2-Tetrachloroethane	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
1,1,2-Trichloroethane	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
1,1-Dichloroethane	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
1,1-Dichloroethene	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
1,2-Dichlorobenzene	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
1,2-Dichloroethane	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
1,2-Dichloropropane	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
1,3-Dichlorobenzene	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
1,4-Dichlorobenzene	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	

KAR Laboratories, Inc.

LABORATORY DETAIL REPORT

Client: **City of Cadillac, Water Resources**

KAR Project No. : **402028**

Date Reported : **04/15/14**

Attest: 
David R. Alkema, Lab Manager

Project

Description: Analysis of one Spring Biosolids sample.

Sample ID : **"Digester #4"**

Sampled By : **DC of City of Cadillac, Water Resources**

Date Received : **04/01/14**

Sample Date : **03/31/14**

Sample Type : **sludge**

Sample Time : **1300**

KAR Sample No. : **402028-01**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
2-Chloroethylvinyl ether	<20,000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Acrolein	<10,000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Acrylonitrile	<4000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Benzene	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Bromodichloromethane	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Bromoform	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Bromomethane	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Carbon tetrachloride	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Chlorobenzene	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Chloroethane	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Chloroform	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Chloromethane	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Cis-1,3-Dichloropropene	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Dibromochloromethane	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Ethylbenzene	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Methylene chloride	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Tetrachloroethene	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Toluene	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Trans-1,2-Dichloroethene	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Trans-1,3-Dichloropropene	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Trichloroethene	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Trichlorofluoromethane	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Vinyl chloride	<2000	ug/kg total solids	EPA 8260	04/02/14	JAR	
Prior. Poll. acids	See below		EPA 8270	04/09/14	GMB	
Prior. Poll. base-neutrals	See below		EPA 8270	04/09/14	GMB	
Prep. SV Acid/BN	Completed		EPA 3545	04/08/14	GMB	
1,2,4-Trichlorobenzene 8270	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
1,2-Dichlorobenzene by 8270	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
1,2-Diphenylhydrazine	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
1,3-Dichlorobenzene by 8270	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
1,4-Dichlorobenzene by 8270	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
2,3,7,8-TCDD by 8270	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
2,4,6-Trichlorophenol	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
2,4-Dichlorophenol	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
2,4-Dimethylphenol	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
2,4-Dinitrophenol	<28,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
2,4-Dinitrotoluene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
2,6-Dinitrotoluene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
2-Chloronaphthalene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
2-Chlorophenol	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
2-Methyl-4,6-dinitrophenol	<28,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
2-Nitrophenol	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
3,3'-Dichlorobenzidine	<28,000	ug/kg total solids	EPA 8270	04/09/14	GMB	

KAR Laboratories, Inc.

LABORATORY DETAIL REPORT

Client: **City of Cadillac, Water Resources**

KAR Project No. : **402028**

Date Reported : **04/15/14**

Attest:


David R. Alkema, Lab Manager

Project

Description: **Analysis of one Spring Biosolids sample.**

Sample ID : **"Digester #4"**

Sampled By : **DC of City of Cadillac, Water Resources**

Date Received : **04/01/14**

Sample Date : **03/31/14**

Sample Type : **sludge**

Sample Time : **1300**

KAR Sample No. : **402028-01**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
4-Bromophenyl phenyl ether	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
4-Chloro-3-methylphenol	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
4-Chlorophenyl phenyl ether	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
4-Nitrophenol	<28,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Acenaphthene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Acenaphthylene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Anthracene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Benzidine	<70,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Benzo(a)anthracene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Benzo(a)pyrene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Benzo(b)fluoranthene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Benzo(ghi)perylene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Benzo(k)fluoranthene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Bis(2-chloroethoxy)methane	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Bis(2-chloroethyl)ether	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Bis(2-chloroisopropyl)ether	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Bis(2-ethylhexyl)phthalate	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Butylbenzyl phthalate	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Chrysene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Di-n-butyl phthalate	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Di-n-octyl phthalate	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Dibenzo(ah)anthracene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Diethyl phthalate	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Dimethyl phthalate	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Fluoranthene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Fluorene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Hexachlorobenzene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Hexachlorobutadiene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Hexachlorocyclopentadiene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Hexachloroethane	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Indeno(123cd)pyrene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Isophorone	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
N-Nitrosodi-n-propylamine	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
N-Nitrosodimethylamine	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
N-Nitrosodiphenylamine	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Naphthalene by Method 8270	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Nitrobenzene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Pentachlorophenol	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Phenanthrene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Phenol	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Pyrene	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Prior. Poll. pest./PCB's	See below		EPA 8270	04/09/14	GMB	
4,4'-DDD	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	

KAR Laboratories, Inc.

LABORATORY DETAIL REPORT

Client: **City of Cadillac, Water Resources**

KAR Project No. : **402028**

Date Reported : **04/15/14**

Attest: 
David R. Alkema, Lab Manager

Project

Description: **Analysis of one Spring Biosolids sample.**

Sample ID : **"Digester #4"**

Sampled By : **DC of City of Cadillac, Water Resources**

Date Received : **04/01/14**

Sample Date : **03/31/14**

Sample Type : **sludge**

Sample Time : **1300**

KAR Sample No. : **402028-01**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
4,4'-DDE	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
4,4'-DDT	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Aldrin	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Alpha-BHC	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Beta-BHC	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Chlordane	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Delta-BHC	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Dieldrin	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Endosulfan I	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Endosulfan II	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Endosulfan sulfate	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Endrin	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Endrin aldehyde	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Gamma-BHC	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Heptachlor	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Heptachlor epoxide	<14,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
PCB Aroclor 1016 by 8270	<140,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
PCB Aroclor 1221 by 8270	<140,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
PCB Aroclor 1232 by 8270	<140,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
PCB Aroclor 1242 by 8270	<140,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
PCB Aroclor 1248 by 8270	<140,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
PCB Aroclor 1254 by 8270	<140,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
PCB Aroclor 1260 by 8270	<140,000	ug/kg total solids	EPA 8270	04/09/14	GMB	
Toxaphene	<140,000	ug/kg total solids	EPA 8270	04/09/14	GMB	

Sample ID : **TCLP Leachate of Digester #4**

Sampled By :

Date Received : **04/01/14**

Sample Date :

Sample Type : **TCLP**

Sample Time :

KAR Sample No. : **402028-01T**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
TC Metals	Completed		EPA 200.7	04/10/14	BPK	
Prep. Hg	Completed		EPA 7470A	04/03/14	BPK	
Prep. metals	Completed		EPA 200.7/8	04/03/14	ZZW	
Arsenic, total, by ICP	<0.5	mg/L	EPA 200.7	04/10/14	BPK	TC regulatory limit is 5.0 mg/L.
Barium, total	1.37	mg/L	EPA 200.7	04/10/14	BPK	TC regulatory limit is 100 mg/L.
Cadmium, total	<0.025	mg/L	EPA 200.7	04/10/14	BPK	TC regulatory limit is 1.0 mg/L.
Chromium, total	<0.05	mg/L	EPA 200.7	04/10/14	BPK	TC regulatory limit is 5.0 mg/L.
Copper, total	<0.1	mg/L	EPA 200.7	04/10/14	BPK	
Lead, total, by ICP	<0.25	mg/L	EPA 200.7	04/10/14	BPK	TC regulatory limit is 5.0 mg/L.

KAR Laboratories, Inc. . .

LABORATORY DETAIL REPORT

Client: **City of Cadillac, Water Resources**

KAR Project No. : **402028**

Date Reported : **04/15/14**

Attest: 
David R. Alkema, Lab Manager

Project

Description: **Analysis of one Spring Biosolids sample.**

Sample ID : **TCLP Leachate of Digester #4**

Sampled By :

Date Received : **04/01/14**

Sample Date :

Sample Type : **TCLP**

Sample Time :

KAR Sample No. : **402028-01T**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Mercury, total	<0.05	mg/L	EPA 7470A	04/03/14	BPK	TC regulatory limit is 0.2 mg/L.
Selenium, total, by ICP	<0.5	mg/L	EPA 200.7	04/10/14	BPK	TC regulatory limit is 1.0 mg/L.
Silver, total	<0.025	mg/L	EPA 200.7	04/04/14	JHB	TC regulatory limit is 5.0 mg/L.
Zinc, total	1.29	mg/L	EPA 200.7	04/10/14	BPK	
TC Semi-Volatiles	See below		EPA 8270	04/03/14	GMB	
Prep, SV Acid/BN	Completed			04/03/14	GMB	
Prep, herbicides	Completed		EPA 8151	04/07/14	GMB	
2,4,5-Trichlorophenol	<0.1	mg/L	EPA 8270	04/03/14	GMB	TC regulatory limit is 400 mg/L.
2,4,6-Trichlorophenol	<0.1	mg/L	EPA 8270	04/03/14	GMB	TC regulatory limit is 2.0 mg/L.
2,4-Dinitrotoluene	<0.1	mg/L	EPA 8270	04/03/14	GMB	TC regulatory limit is 0.13 mg/L.
Cresols	<0.2	mg/L	EPA 8270	04/03/14	GMB	TC regulatory limit is 200 mg/L.
Hexachlorobenzene	<0.1	mg/L	EPA 8270	04/03/14	GMB	TC regulatory limit is 0.13 mg/L.
Hexachlorobutadiene	<0.1	mg/L	EPA 8270	04/03/14	GMB	TC regulatory limit is 0.5 mg/L.
Hexachloroethane	<0.1	mg/L	EPA 8270	04/03/14	GMB	TC regulatory limit is 3.0 mg/L.
Nitrobenzene	<0.1	mg/L	EPA 8270	04/03/14	GMB	TC regulatory limit is 2.0 mg/L.
Pentachlorophenol	<0.1	mg/L	EPA 8270	04/03/14	GMB	TC regulatory limit is 100 mg/L.
Pyridine	<0.2	mg/L	EPA 8270	04/03/14	GMB	TC regulatory limit is 5.0 mg/L.
TC Pesticides by ECD	See below		EPA 8081	04/04/14	GMB	
Prep, ECD	Completed		EPA 3510	04/03/14	GMB	
Chlordane	<0.02	mg/L	EPA 8081	04/04/14	GMB	TC regulatory limit is 0.03 mg/L.
Endrin	<0.004	mg/L	EPA 8081	04/04/14	GMB	TC regulatory limit is 0.02 mg/L.
Heptachlor	<0.004	mg/L	EPA 8081	04/04/14	GMB	TC regulatory limit is 0.008 mg/L.
Heptachlor epoxide	<0.004	mg/L	EPA 8081	04/04/14	GMB	TC regulatory limit is 0.008 mg/L.
Lindane by ECD	<0.004	mg/L	EPA 8081	04/04/14	GMB	TC regulatory limit is 0.4 mg/L.
Methoxychlor	<0.02	mg/L	EPA 8081	04/04/14	GMB	TC regulatory limit is 10.0 mg/L.
Toxaphene	<0.04	mg/L	EPA 8081	04/04/14	GMB	TC regulatory limit is 0.5 mg/L.
TC Herbicides	See below		EPA 8151	04/08/14	GMB	
2,4,5-TP (Silvex)	<0.2	mg/L	EPA 8151	04/08/14	GMB	TC regulatory limit is 1.0 mg/L.
2,4-D	<0.2	mg/L	EPA 8151	04/08/14	GMB	TC regulatory limit is 10.0 mg/L.

Sample ID : **ZHE Leachate of Digester #4**

Sampled By :

Date Received : **04/01/14**

Sample Date :

Sample Type : **ZHE**

Sample Time :

KAR Sample No. : **402028-01Z**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
TC Volatiles	See below		EPA 8260	04/03/14	JAR	
Prep, VOA	Completed		EPA 5030	04/03/14	JAR	
1,1-Dichloroethene	<0.1	mg/L	EPA 8260	04/03/14	JAR	TC regulatory limit is 0.7 mg/L.
1,2-Dichloroethane	<0.1	mg/L	EPA 8260	04/03/14	JAR	TC regulatory limit is 0.5 mg/L.

KAR Laboratories, Inc.

LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : **402028**

Date Reported : **04/15/14**

Attest: 
David R. Alkema, Lab Manager

Project

Description: *Analysis of one Spring Biosolids sample.*

Sample ID : **ZHE Leachate of Digester #4**

Sampled By :

Date Received : **04/01/14**

Sample Date :

Sample Type : **ZHE**

Sample Time :

KAR Sample No. : **402028-01Z**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
1,4-Dichlorobenzene	<0.1	mg/L	EPA 8260	04/03/14	JAR	TC regulatory limit is 7.5 mg/L.
Benzene	<0.1	mg/L	EPA 8260	04/03/14	JAR	TC regulatory limit is 0.5 mg/L.
Carbon tetrachloride	<0.1	mg/L	EPA 8260	04/03/14	JAR	TC regulatory limit is 0.5 mg/L.
Chlorobenzene	<0.1	mg/L	EPA 8260	04/03/14	JAR	TC regulatory limit is 100 mg/L.
Chloroform	<0.1	mg/L	EPA 8260	04/03/14	JAR	TC regulatory limit is 6.0 mg/L.
Methyl ethyl ketone	<2	mg/L	EPA 8260	04/03/14	JAR	TC regulatory limit is 200 mg/L.
Tetrachloroethene	<0.1	mg/L	EPA 8260	04/03/14	JAR	TC regulatory limit is 0.7 mg/L.
Trichloroethene	<0.1	mg/L	EPA 8260	04/03/14	JAR	TC regulatory limit is 0.5 mg/L.
Vinyl chloride	<0.1	mg/L	EPA 8260	04/03/14	JAR	TC regulatory limit is 0.2 mg/L.